The What, When, Whether Model of Intentional Action

MARCEL BRASS and PATRICK HAGGARD

The question of how we can intentionally control our behavior has an enduring fascination for philosophers, psychologists, and neurologists. Brain imaging techniques such as functional MRI have recently provided new insights into the functional and brain mechanisms involved in intentional action. However, the literature is rather contradictory and does not reveal a consistent picture of the functional neuroanatomy of intentional action. Here the authors argue that this confusion arises partly because intentional action has been treated as a unitary concept within neuroscience, even though experimental studies may focus on any of a number of different aspects of intentional action. To provide a heuristic framework for the investigation of intentional action, the authors propose a model that distinguishes three major components: a component related to the decision about which action to execute (*what* component), a component that is related to the decision about when to execute an action (*when* component), and finally the decision about whether to execute an action or not (*whether* component). Based on this distinction, the authors review some key findings on intentional action and provide neuroscientific evidence for the What, When, Whether (WWW) model of intentional action. NEUROSCIENTIST 14(4):319–325, 2008. DOI: 10.1177/1073858408317417

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Our behavior is based on interactions between the environment and our own intentions and so can be understood as a balanced interplay of contextual information and internal factors. The importance of this balance is dramatically illustrated by the behavioral consequences of patients with so-called anarchic hand syndrome or alien hand syndrome (Della Sala and others 1991). Such patients are not able to intentionally control the actions of one hand. Consequently, the "anarchic hand" continually responds to environmental stimuli, even when the patient does not want to perform any actions at all. Moreover, the anarchic hand often interferes with the actions that the patient does want to perform with the "good" hand. Although performance research has long primarily focused on external determinants of motor performance using a stimulus-response (S-R) approach, recent research has begun to address the question of the functional and neural correlates of intentional control of action. On the brain level, motor control is primarily a domain of the frontal cortex (in interaction with the parietal cortex). More specifically, it has been suggested that the fronto-median cortex is involved in the intentional control of behavior whereas the fronto-lateral cortex is involved in external control (Goldberg 1985). Functional brain imaging research of the past 15 years has refined this picture while retaining this broad distinction. Different areas of the fronto-median wall have been related to intentional action (Lau and others 2004b; Walton and others 2004; Nachev and others 2005; Lau and others 2006; van Eimeren and others 2006). However, the literature is rather contradictory and does not reveal a consistent picture of the specific brain areas that are involved in intentional action. This confusion arises partly because intentional action has been treated as a unitary concept within neuroscience, even though experimental studies may focus on any of a number of different aspects of intentional action.

Therefore, understanding the brain basis of intentional action requires understanding what intentional action is. The most widely accepted property of intentional actions is that they are somehow both purposive and endogenous: the action comes from, or is made by, an agent, rather than an external cause. Intentional actions can therefore be defined by contrasting them with reflex, or purely stimulus-driven actions. In reflex actions, an external stimulus causes immediate, stereotyped motor behavior. In intentional actions, by contrast, there is no obvious external stimulus. Motor behavior may be flexible in form and timing, yet still be related to purpose.

Therefore, the nervous system must make decisions, or generate additional information, to produce intentional behaviors, which is not required for stimulus-driven action. Cognitive analysis of the different kinds of information generated helps to understand the different

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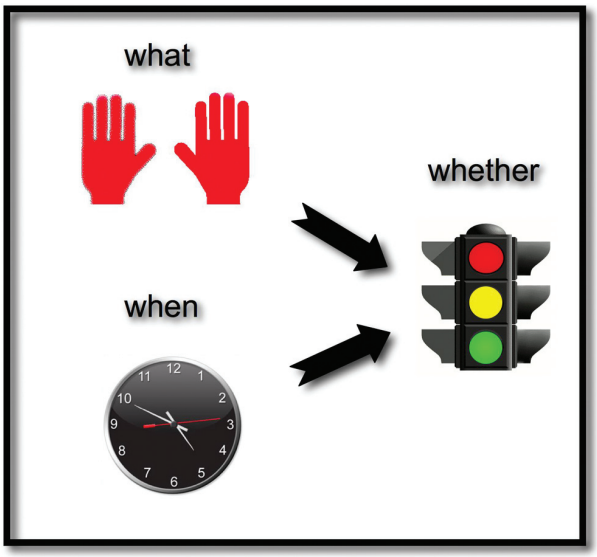

Fig. 1. Schematic drawing of the What, When, Whether (WWW) model of intentional action. The model assumes three component decisions relevant to intentional action. 1. The *what* component related to the decision of which action to execute. 2. The *when* component reflecting the decision of when to execute an action. 3. The *whether* component related to the decision of whether to execute an action or not.

neural systems and processes of intentional action. We suggest that intentional action depends on at least three key kinds of information-generation: deciding what to do, deciding when to do it, and deciding whether to implement one's decision or not (Fig. 1). The *what* component of intentional action reflects the decision of which action to execute (Lau and others 2004b; Walton and others 2004; Cunnington and others 2006; van Eimeren and others 2006; Mueller and others 2007). It is often contrasted with perceptual decision making in stimulus-response association tasks. The *when* component has received most attention in the literature on intentional action (Libet and others 1983; Cunnington and others 2002; Cunnington and others 2003; Lau and others 2004a), perhaps because it leads naturally to questions about free will and the causation of intentional actions. Finally, we think that a third component of decision should also be considered, despite being so far widely ignored. We call this the *whether* component (Brass and Haggard 2007) because it determines whether to execute any intentional action or not. We now review these processes in turn and show that they depend on distinct neural processes, occurring in different regions of the brain.

Action Selection: The *What* Component of Intentional Action

The first crucial aspect of voluntary action relates to the decision of which action to execute. Only a few studies have investigated the *what* component of intentional action directly. This might be due to severe methodological and theoretical problems that arise from investigating intentional selection of action (e.g., Lau and others 2004a). Commonly the what component is tested in paradigms in which participants can freely choose between different action alternatives (Lau and others 2004b; Walton and others 2004; Cunnington and others 2006; van Eimeren and others 2006; Mueller and others 2007).

When comparing brain activity in a condition that requires the free selection of action with a condition where the responses are externally triggered, activity in different parts of the fronto-median wall has been reported (Fig. 2). The most consistent fronto-median brain activity was located in the rostral cingulate zone (RCZ) and the pre-SMA (Lau and others 2004b; Walton and others 2004; Mueller and others 2007). Interpreting such activity as the neural correlate of intentional selection of action has been complicated by two alternative interpretations: First, fronto-median activity in such paradigms has been linked to conflict resolution rather than intentional processes (Botvinick and others 2001; Nachev and others 2005). On this view, the activation might be explained by the conflict between alternatives, rather than by deciding which to select. Second, such tasks may not capture action selection in its natural form because of a methodological shortcoming. Specifically, participants are instructed to choose but are not told what to choose or how to choose it. They are often instructed to avoid selecting actions by rules (e.g., not always pressing the same key or not alternating between keys). Such instructions might lead to the attempt of participants to generate a random sequence of responses (Jahanshahi and Dirnberger 1999). These studies would then investigate random number generation rather than intentional selection. However, recent studies on intentional action randomly intermixed free selection trials with trials where the response was externally triggered, making this strategy less likely (Lau and others 2004b; van Eimeren and others 2006; Mueller and others 2007).

A crucial aspect of the what component of voluntary action is the competition between different response alternatives (Botvinick and others 2001; Nachev and others 2007). To decide for a specific behavior, one has to overcome conflict from competing response alternatives. Such competition is stronger when the action is not externally triggered, because in a free selection situation, all potential response alternatives have a rather similar activation level. Therefore it has been suggested that activity in the fronto-median cortex in intentional selection paradigms in fact reflects competition between different response alternatives rather than intentional selection of action. In accordance with this assumption, several studies relate activity in the fronto-median wall to conflict processing (for an overview, see Ridderinkhof and others 2004; Rushworth and others 2004). In a series of recent studies with neurological patients lesioned in the caudo-dorsal part of the fronto-median wall, Sumner and colleagues (2007) substantiated this claim. They demonstrated that a patient with a specific SMA lesion was not able to suppress unconscious competing manual responses. Nachev and colleagues argue that

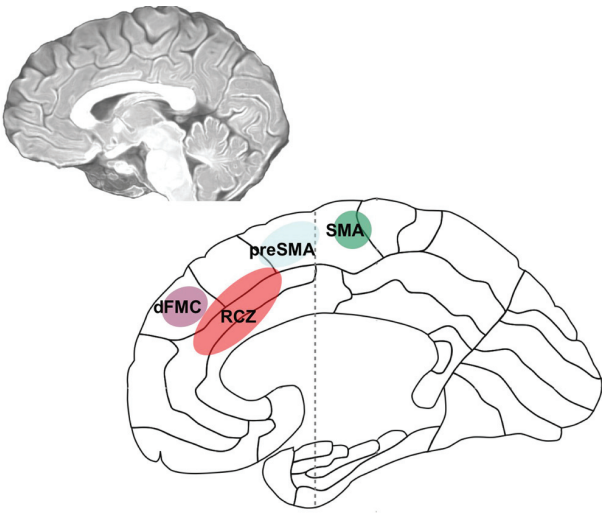

Fig. 2. *Upper part*, Median view of the human cortex. *Lower part*, Schematic drawing of the frontal brain regions that have been consistently found to be involved in the when, what, and whether components of intentional action. SMA = supplementary motor area; pre-SMA = presupplementary motor area; RCZ = rostral cingulate zone; dFMC = dorsal fronto-median cortex.

the SMA/SEF complex is responsible for the automatic inhibition of competing response alternatives. They attribute an inhibitory function to the pre-SMA as well (Nachev and others 2007). Two recent studies tried to directly compare brain activity related to response conflict and intentional selection of action. Lau and colleagues (2006) found a dissociation of both experimental manipulations, with the RCZ being more active for response conflict and the pre-SMA being more active for intentional selection. In contrast, Nachev and colleagues (2005) found a dissociation within the pre-SMA with the more rostral part being related to resolving response conflict and a more caudal part being related to free selection.

At the moment, this controversy is not settled. However, it might be an artificial argument, because intentional action and response conflict may effectively be two sides of the same coin. William James (1890, p. 1134) already recognized that "every mental representation of a movement awakens to some degree the actual movement which is its object; and awakens it in a maximum degree whenever it is not kept from so doing by an antagonistic representation present simultaneously to the mind." From this perspective, response conflict is an inherent property of all action and intentional selection is necessarily required in such situations.

Another promising line of research involves dissociating the neural basis of the when component and what component of intentional action. In a recent study, Mueller and colleagues (submitted) independently manipulated the when and the what components. They found the RCZ to be related to the what component whereas the when component was related to activity in the superior frontal gyrus. These results suggest a basic dissociation between the when and the what components of intentional action. However, the existing empirical data regarding the localization of these different components of intentional action are still contradictory. Whereas some studies point to the crucial role of the RCZ in intentional selection of action (Walton and others 2004; Mueller and others 2007; Mueller and others submitted), others suggest that the pre-SMA is most crucial (Lau and others 2006).

In the richer landscape of intentional behaviors outside the laboratory, the what component may relate to optimizing how to achieve a specific purpose. In contrast, the when component may reflect the fact that people generally have several concurrent purposes but can only perform intentional actions one at a time, and in series. Thus, *what* decisions have a winner-takes-all quality, but *when* decisions involve executive functions such as scheduling and prioritization.

When to Go? Timing of Intentional Action

Internal timing seems central to the concept of intentional action. Reflex actions occur immediately after a stimulus, by definition. In contrast, intentional actions should occur at more random times. This has sometimes led to searches for an internal trigger causing such actions, as if from a homunculus, or ghost in the machine. Modern neuroscience rejects such dualism. Instead, intentional actions may be characterized by their looser connections to external events compared to stimulus-driven actions, by their relations to complex contextual patterns rather than identifiable single stimuli, and by their mediation via memory traces rather than immediate stimulation. Nevertheless, the onset time of brain processes culminating in intentional action has proved important in understanding the flexibility and genesis of action, as well as providing a neuroscientific perspective on "free will."

A key first step came from the technical advances of back-averaging EEG data. This allowed Kornhuber and Deecke (1965) to identify a gradual increase in negativity over frontal motor areas beginning some 2 s before self-paced intentional actions. High-resolution recordings suggest that this readiness potential (RP) begins in the pre-SMA (Ball and others 1999; Cunnington and others 2003) and is followed by activity in the motor areas contralateral to the body part selected for movement. Strong evidence that the RP is indeed involved in the generative process of intentional action came from Jahanshahi and others' (1995) finding that readiness potentials were reduced or absent when movements followed sensory instructional cues.

In addition to the objective absence of an immediate stimulus, intentional actions are associated with an experience of endogenously initiating action. This perhaps explains the widely held view that "I" control my actions, which is central to human culture, and notably to legal systems. Libet and others (1983) investigated this view by comparing the temporal order of RP onset

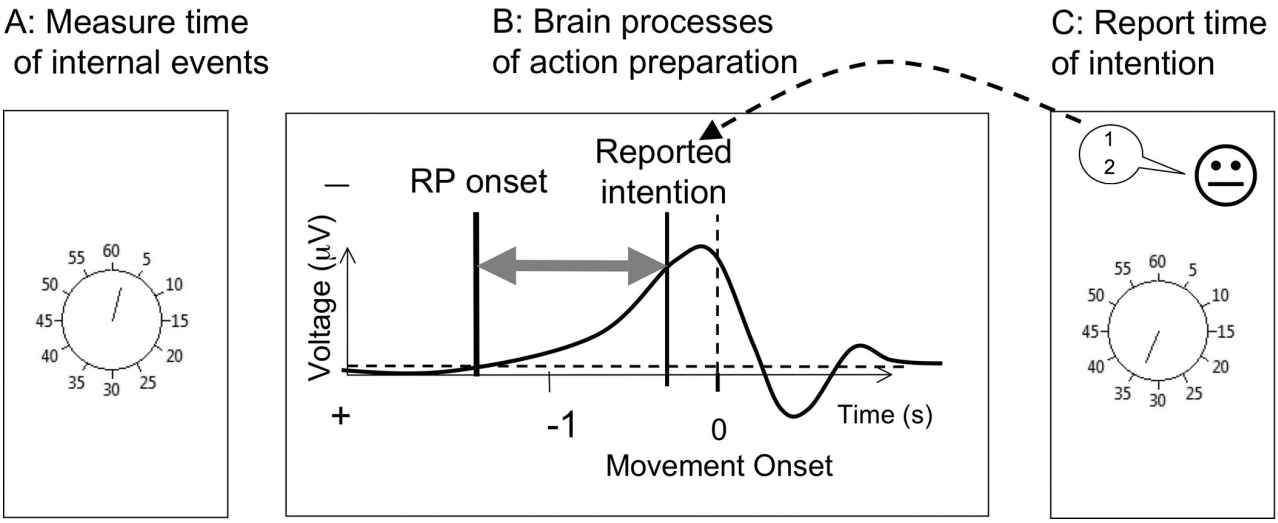

Fig. 3. Libet and others' (1983) method for reporting the time of conscious intentions. Subjects watch a rotating clock (*A*) and make movements at a time of their own choosing. The readiness potential (RP) prior to movement is measured from scalp electrodes over the frontal motor areas (*B*). After the movement, the subject reports the clock position where they first felt the intention to move (*C*). This time is found to occur significantly later than the onset of the RP in *B* (horizontal gray arrow), thus ruling out the interpretation that the subject's conscious intention causes the brain activity that produced the action.

and the conscious experience of intention. Participants made a hand movement at a time of their own choice, while watching a spot slowly rotating around a clock face (Fig. 3). At a random interval after they moved, the spot stopped, and they judged the position it had occupied when they first intended to make the action. This moment of conscious intention averaged 206 ms prior to movement onset. In contrast, the RP began much earlier, around 1000 to 500 ms prior to movement onset. Because conscious intention followed the RP that culminated in the action, Libet argued, conscious thought cannot cause the RP. Rather, the RP must cause both the movement and the conscious experience of being about to move.

The method of Libet's experiment has been repeatedly criticized, although the basic result has been replicated (e.g., Haggard and Eimer 1999). Two essential criticisms dominate. First, although the clock may not be independent of the action itself, participants may react to the spot being in particular clock positions, rather than generate truly intentional actions. Second, the inference depends on accepting that a bona fide mental state occurred at the time subsequently reported by participants using the clock. Yet human temporal judgment often shows substantial biases, including retrospective "backwards referral" of experiences to earlier points in time. Such methodological problems have led to skepticism whether intentions are a valid concept for neuroscience at all. However, other methods that avoid these problems suggest that a conscious experience related to impending intentional action may exist. Fried and others (1991) stimulated the SMA directly with intracortical electrodes prior to neurosurgery for intractable epilepsy. Stimulation at low current at these sites produced an "urge to move" a specific body part. Stronger stimulation produced actual movements, normally of the same limb previously linked to the urge. Clearly, the cortical activity induced by stimulation may differ from natural function. However, these results show that an experience that seems to be related to intention arises as part of the processes that lead to movement. The urge reported at low current cannot reflect an indirect inference or reconstruction by the participant, because they had not yet experienced any physical movement requiring explanation. Fried's result raises several interesting questions, which may be addressed by future research. The first is a more detailed psychological description of the experience of urge, including the time delay between onset of stimulation and awareness. Second, EMG recording could be used to confirm that the urge is indeed of central origin, rather than peripheral reafference from slight muscle contractions.

Although Fried and others stimulated the SMA, the experiences their participants reported may have been generated in other areas remotely activated due to their connections with SMA. Indeed, it remains unclear which brain areas underlie the experience of intentional action. Sirigu and others (2004) measured the perceived time of conscious intention in a group of patients with focal parietal lesions. They found that parietal patients showed much less the anticipatory awareness of intention than a healthy control group, or a group with cerebellar lesions. Indeed, the parietal patients reported intentions to move only briefly before keypress actions, by which time their hand muscles were presumably already active. This deficit could not be attributed to poor time estimation in general, because estimates of the time of the keypress itself were similar in all groups. This result suggests that the parietal cortex may play a key role in conscious monitoring of the preparation of intentional action by the frontal lobes.

Lau and others (2004b) replicated Libet and others' (1983) task of estimating the time of conscious intention in an fMRI scanner. They compared these activations to a

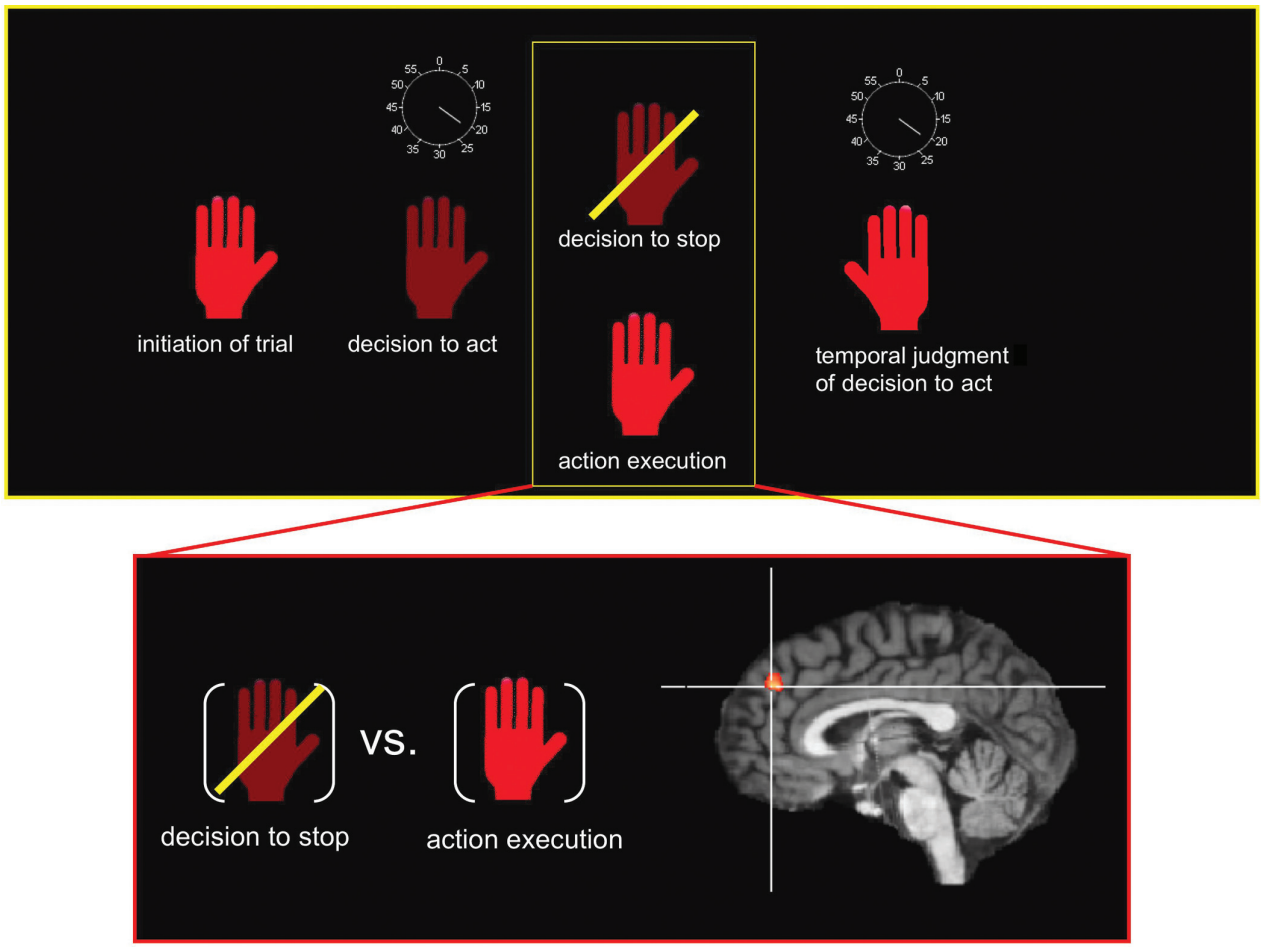

Fig. 4. *Upper part*, Schematic drawing of the intentional inhibition paradigm by Brass and Haggard (2007). Participants could freely decide when to execute an action and were instructed to remember the time when they decided to execute the action. In some trials, they should freely decide to inhibit the planned action in the last moment. After executing or inhibiting the action, they had to indicate when they decided to execute the action. *Lower part*, Brain activity in the dorsal fronto-median cortex when participants decided to inhibit the planned action compared to the execution of the action.

control task in which subjects estimated the time of the keypress action itself. Judging intention led to greater activity in pre-SMA, the intraparietal sulcus, and dorsolateral prefrontal cortex relative to judging action. The combined contribution of pre-SMA and the parietal cortex, taken together with Fried and others' and Sirigu and others' results, suggests that intentional action is not generated by a single brain area but is a product of a recurrent fronto-parietal network. Moreover, because patients with basal ganglia degeneration often show poverty of intentional action, together with reduced readiness potentials (Jahanshahi and others 1995), this network may act as a cognitive-motor loop transforming contextual information from the parietal cortex through the basal ganglia into a drive to frontal motor areas.

To Do or Not to Do: The *Whether* Component of Intentional Action

Although the when and the what components of intentional action have received increasing attention during the past few years, the intentional inhibition of action has been widely neglected. Most research on inhibition has focused on inhibiting actions elicited by external stimuli (Aron and others 2004). For example, an imperative stimulus is presented together with a further stimulus instructing the participant to withhold the action (Logan and others 1984). However, in our daily life, we very often have to decide ourselves whether we should act or not. Furthermore, overcoming impulsive behavior is crucial for many cooperative social interactions. In a recent fMRI experiment, we (Brass and Haggard 2007) started to address the question of which brain areas are related to the intentional inhibition of action. Similar to the classical Libet paradigm, participants had to decide when to press a key and were to determine on a clock the time when they decided to press. In addition, they were instructed to sometimes withhold the keypress in the last moment (Fig. 4). In contrast to classical stop signal paradigms, participants could decide themselves whether to execute the action or not. When comparing intentional inhibition of action with the execution of action, we found activity in the dorso-fronto-median cortex (Fig. 4) and the

anterior insula. Interestingly, the fronto-median activation was located anterior to the pre-SMA and dorsal to the rostral cingulate zone (Fig. 2). These data support the idea that the *whether* component of intentional action can be distinguished from the when and what components. Furthermore, our data suggest that a specific area in the fronto-median cortex is related to a kind of self-control. This point was further corroborated by a recent study on gambling that also investigated the intentional inhibition of action (Campbell-Meiklejohn and others 2008). A very strong behavioral tendency of pathological and nonpathological gamblers is to continue gambling to recover losses (loss-chasing). In this study, quitting loss chasing behavior activated a brain region that overlapped with the dorsal fronto-median cortex we found. As in Brass and Haggard, participants had to decide themselves whether to stop an action or not. Both studies identify a specific fronto-median brain area to be related to the whether component of intentional action. Future research has to show whether this brain region is crucial for self-control in other domains such as addiction or aggressive behavior.

The possibility to prepare but then inhibit intentional actions at the last moment may be an important evolved feature of human cognition. For example, it could allow an action to be simulated but not actually performed. This may be particularly important in social settings, where the immediate short-term gain of the individual is balanced against longer-term gains accruing from consideration of others.

Conclusions

We suggest that intentional actions involve several decisions that are absent from stimulus-driven actions. These include deciding what action to perform and when to perform it. We also suggest intentional actions may involve a final check whether the action should be performed or not. We call these the *what*, *when*, and *whether* components of intentional action, respectively. By highlighting the brain bases of each decision, and the relation between them in a common framework (the WWW model, Fig. 1), we hope to guide experimental work on intentional action, but also to gain a deeper understanding of pathologies affecting intentional behavior. The anarchic hand syndrome and utilization behavior, for example, might reflect the inability to intentionally select behavior (what component). In contrast, the difficulties of movement initiation in Parkinson's disease suggest an impairment of the when component. Finally, obsessive-compulsive behavior, Tourette's syndrome, and ADHD might involve inability to intentionally inhibit actions (whether component).

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